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Sol–gel nanoglues for an organic binder-free TiO_2 nanofiber anode for lithium ion batteries†

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A TiO_2 nanofiber anode is glued using a sol–gel reaction to improve cycle performance and Coulombic efficiencies. The hydrolysis-condensation process produces TiO_2 nanoglues consisting of TiO_2 nanoparticles, providing strong adhesion between nanofibers and the current collector as well as between nanofibers, rendering favourable transporting properties of electrons and Li ions in the TiO_2 nanofiber electrode, and eventually leading to the excellent performance of Li ion batteries exhibiting an excellent initial Coulombic efficiency (~85%), cycle performance up to 100 cycles, and high rate capability (60% at 5 C).

TiO_2 has been considered as an attractive material for use in a lithium ion battery anode due to its low cost and excellent structural stability during lithiation–delithiation processes.[1,2] However, TiO_2 has disadvantages of low Li ion diffusivity and poor electrical conductivity, which limit its practical application.[3] To overcome these problems, 1D nanostructures such as nanowires, nanofibers, and nanotubes are adopted because they provide efficient electron pathways and short diffusion length for lithium ions.[3–6] Also, surface modification of electrode materials[5] and the use of hybrid electrode systems[7] have been proven to be effective ways for improved Li ion diffusivity and electrical conductivity. In a different way, the structure of electrode materials can affect Li ion diffusivity and electrical conductivity.[8] The formation of a direct electrical contact between electrode materials and the current collector improved the electrochemical performance of the 1D nanostructured TiO_2 electrode.[8]

A recent investigation showed that adhesion between electrode materials and the current collector as well as between the materials can affect the electrochemical performance of Li ion batteries.[9] In general, to provide strong adhesion between these components, an electrode is generally prepared by mixing active materials with polymeric binders. Conducting additives such as carbon black and graphene are also added to compensate for low electrical conductivity of active materials.[10] However, the use of large amounts of processing additives decreases volumetric capacity of batteries because those are generally electrochemically inactive materials.[11] In addition, typical polymer binders are insulating materials therefore they can block conducting pathways between electrode materials and the current collector as well as between electrode materials.

Here, we develop a novel method for producing Li ion batteries with stable cycle performance without the addition of processing additives that decrease volumetric capacity and hinder the electrochemical reaction. We demonstrate the fabrication of a 1D nanostructured TiO_2 anode by using this fabrication method. To fabricate the 1D nanostructure, electrospinning is utilized in this study because it is an easy, cost-effective, and versatile methodology for generating 1D nanostructures of various kinds of materials.[12] The method proposed here utilizes a general sol–gel route that is the hydrolysis–condensation process of metal alkoxide. During the sol–gel reaction, titanium alkoxide ($Ti(OR)_4$) forms titanium oxide networks (–Ti–O–Ti–) between the nanofiber structures.[13] The calcination process leads to the formation of particle-like metal oxides, which binds chemically each nanofiber as well as adheres nanofibers on the current collector. In this work, we call this unique structure "TiO_2 sol–gel nanoglues". The nanoglues between nanofibers provide additional electron conducting pathways between nanofibers and the current collector as well as play a role in binding active materials and adhering them on the current collector like polymeric binders. At the same time, it can avoid the decrease in specific capacity and volumetric capacity because the nanoglues consist of electrochemically active materials, in this experiment TiO_2. The glues can fill gap spaces between the nanofibers, strengthening contact properties of the whole electrode, and consequently lead to the stable cycle performance with high initial Coulombic efficiency. The

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rate capability of a TiO_2 nanofiber electrode with TiO_2 sol–gel nanoglues is also evaluated. The electrode exhibits highly stable rate capability as a function of cycle number, maintaining an excellent rate capability as high as a TiO_2 nanofiber electrode. These electrochemical performances prove that the nanoglue system promotes adhesion between electrode materials and the current collector without adding polymeric binders, and it also provides constructive conducting pathways, leading to the excellent stability in cycle performance. Such excellent cycle performance is also observed even under the fast charging–discharging conditions.

The processing steps for a TiO_2 nanofiber electrode strengthened with TiO_2 sol–gel nanoglues are schematically depicted in Fig. 1. First, 1 g of poly (vinyl acetate) (PVAc, $M_w \sim$ 500 000, Sigma-Aldrich) was dissolved in 5 mL of *N*,*N*-dimethylformamide (DMF, 99.8%, Sigma-Aldrich). 0.75 g of titanium(IV) isopropoxide (97%, Sigma-Aldrich) and 0.3 g of acetic acid (99.5%, Samchun Chemical) were added to the PVAc stock solution, followed by stirring for at least 6 h. The prepared solution was transferred to a syringe equipped with a metal nozzle (18 G). Electrospinning was carried out on a stainless steel foil (SUS 304, Nialco) (step 1). An electrical voltage of 15 kV was applied at a distance of 15 cm. The electrospun nanofibers were hydrolysed for 6 h under ambient conditions. After annealing at 450 °C for 3 h, the TiO_2 nanofibers were obtained (step 2). TiO_2 sol–gel nanoglues were prepared using the same precursor solution for TiO_2 nanofibers. The prepared precursor solution was spin coated on the preformed TiO_2 nanofibers at 4000 rpm for 30 s (step 3). After hydrolysis followed by annealing, a TiO_2 nanofiber electrode strengthened by the nanoglues was obtained (step 4). The resulting loading mass of TiO_2 nanofibers and TiO_2 nanofibers strengthened with nanoglues was 33.23 and 368.3 μg, respectively. The mass ratio of TiO_2 nanofibers to nanoglues in the electrode was ~0.1.

Fig. 2 shows the morphological information of the TiO_2 electrodes. A thin film of the nanoglues was formed on the substrate by spin-coating followed by annealing, and the morphology was observed by scanning electron microscopy (SEM). Fig. 2a shows

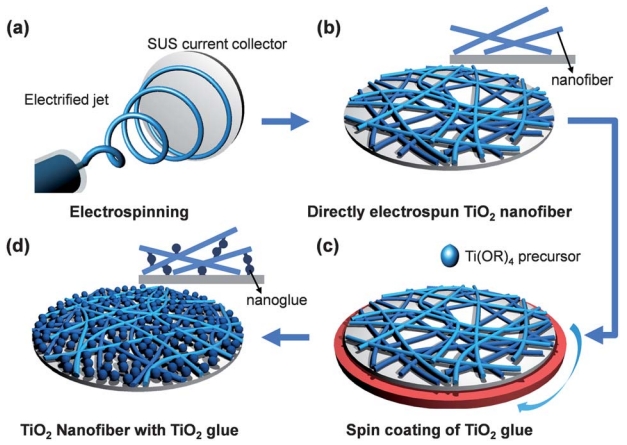

Fig. 1 Schematic illustration of processing steps for a TiO_2 nanofiber electrode modified with TiO_2 nanoglue.

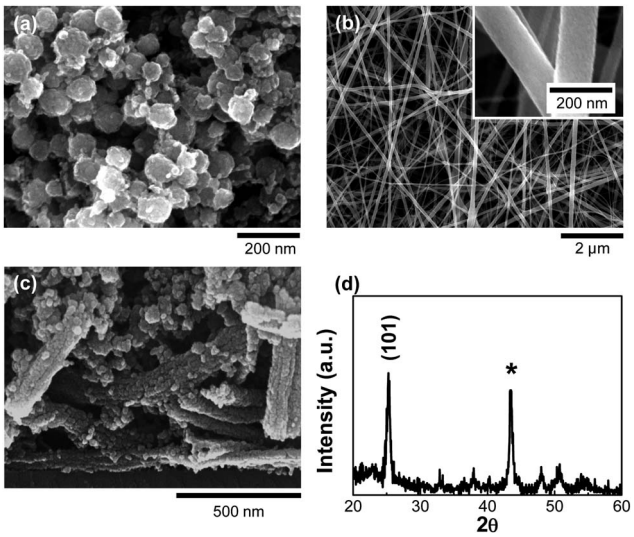

Fig. 2 Scanning electron microscopy (SEM) images of (a) TiO_2 glue, (b) TiO_2 nanofiber. (c) Cross-sectional SEM image of TiO_2 nanofiber connected with TiO_2 glue, and (d) XRD patterns of the TiO_2 glue and TiO_2 nanofiber.

that TiO_2 sol–gel nanoglues consist of both large nanoparticles of ~100 nm and fine ones under 50 nm. Fig. 2b shows the SEM image of the directly electrospun TiO_2 nanofibers on the current collector. The nanofibers have an average diameter of ~100 nm with a continuous 1D structure. The cross-sectional SEM image (Fig. 2c) of TiO_2 nanofiber strengthened with nanoglues shows that the nanoparticles are homogeneously coated on the surface of the preformed TiO_2 nanofibers and fill the spaces between TiO_2 nanofibers and the current collector. Therefore, the nanoglues contribute to increase of the contact area of electrode materials on the current collector as well as connecting nanofibers. Fig. 2d shows the X-ray diffraction (XRD) pattern of TiO_2 nanofibers with TiO_2 nanoglues. There are two obvious peaks at 25.25 and 43.40°. The peak observed at lower degree can be assigned to the (101) plane of anatase TiO_2 (JCPDS 21-1272). There are no any traceable peaks corresponding to other crystal forms of TiO_2 such as brookite and rutile. The peak observed at higher degree comes from the SUS current collector. Also, we compare the XRD patterns of TiO_2 nanostructures as shown in Fig. S1 (see ESI†) The XRD patterns show that all TiO_2 nanostructures have the same crystal structure of anatase and the crystal structures of TiO_2 nanofibers were not altered by the application of TiO_2 sol–gel nanoglues.

To show the binder-like function of TiO_2 sol–gel nanoglues, the adhesion tests were conducted and the results are shown in Fig. S2 (see ESI†). In general, the Scotch tape method is utilized to observe the adhesion properties of a layer on a substrate.[14] In this study, a poly(dimethylsiloxane) (PDMS) elastomeric stamp is used to apply the pressure homogeneously over the sample. The samples were contacted with a PDMS stamp (prepolymer : curing agent = 10 : 1 (w/w); SYLGARD 184 silicone elastomer kit, Dow Corning) and vertical loading was applied by placing different balance weights of 50 g, 200 g, and 1 kg. After 1 min, PDMS was detached from the samples. The initial and residual weights for TiO_2 on the SUS foil were measured using a

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microbalance (Sartorius SE2, resolution 0.1 μg, Sartorius, Germany). In the case of nanofibers, a great deal of nanofibers is detached, while the nanofibers strengthened with nanoglues strongly adhere to the substrate. The adhesion test reveals that the nanoglues reinforce the adhesion strength of TiO_2 nanofiber.

To investigate the influence of nanoglues on the electrochemical performance of a TiO_2 nanofiber electrode, two types of TiO_2 nanofiber electrodes were prepared according to the presence of nanoglues. Both electrodes were assembled in an argon filled glove box. The nanostructured TiO_2 electrode was used as a working electrode and pure lithium foil was used as both reference and counter electrodes. 1.0 M $LiPF_6$ in ethylene carbonate-diethylene carbonate (EC/DEC, 1 : 1 vol%) was used as an electrolyte. The electrochemical performance of the electrodes was measured using a battery cycle tester TOSCAT 3100 (Toyo Systems). The electrodes were galvanostatically discharged and charged in a voltage range 1–3 V *vs.* Li/Li^+ at the current rate of 0.2 C (1 C = 200 mA g^{-1}).

Fig. 3a compares the curves for the first discharge and charge capacities of two different TiO_2 electrodes. All electrodes show two plateaus at 1.77 and 1.89 V in the discharge and charge curves, respectively, which correspond to the phase change between TiO_2 and lithiated TiO_2 by the lithium insertion/extraction.[15–19] We confirmed these reaction voltages by cyclic voltammetry (CV). The CV curves were obtained in the voltage range between 3.0 V and 1.0 V at a scan rate of 0.2 mV s^{-1}.[17] The results are shown in Fig. S3 (see ESI†). Both electrodes exhibited a cathodic peak at 1.7 V and an anodic peak at 2.0 V, which corresponds well to the previously reported values for anatase TiO_2.[18,19] The lithiation and delithiation processes for anatase TiO_2 can be expressed as the following equation:

$$TiO_2 + xLi^+ + xe^- \leftrightarrow Li_xTiO_2$$

where x is the amount of inserted Li ions (mole). Anatase TiO_2 has a theoretical capacity of 335 mA h g^{-1} ($x = 1$).[20] The TiO_2 nanofiber electrode exhibits the first discharge and charge capacities of 351 ($x = 1.05$) and 193 mA h g^{-1} ($x = 0.57$), respectively. The capacity value of the first discharge is larger than theoretical capacity. A similar result was reported elsewhere[15] which may be due to the side effect induced by the interaction with an electrolyte.[21] The electrode made of TiO_2 nanofibers strengthened with nanoglues delivers discharge and charge capacities of 204 ($x = 0.61$) and 174 mA h g^{-1} ($x = 0.52$), respectively. Fig. 3b shows the capacity retention of both electrodes as a function of cycle number up to 100 cycles; the values of capacity retention were obtained from the cycle performance results shown in Fig. S4 (see ESI†). For the TiO_2 nanofiber electrode, the capacity decreases steeply until 5 cycles and after that it decreases slowly, while the cycle performance is very stable for TiO_2 nanofiber with nanoglues over the whole measured cycles. Also, the initial Coulombic efficiency for the electrode composed of TiO_2 nanofibers and nanoglues is 84.7% and the efficiency reaches a level >99% within 10 cycles (Fig. 3c). In contrast , the TiO_2 nanofiber electrode shows an initial Coulombic efficiency of only 55% and the curve is fluctuated. Based on the results, it can be concluded that the presence of TiO_2 nanoglues in the nanofiber electrode leads to the improved cycle performance and higher initial Coulombic efficiency. The high initial Coulombic efficiency observed here would be an advantage in the fabrication of a full cell that is practical application.

The rate capability of the electrodes was also measured with increasing current density from 0.2 C to 5 C as shown in Fig. 3d; the values of capacity retention were obtained from the rate capability results shown in Fig. S5 (see ESI†). The capacity retention was calculated using each capacity value of the first cycle at 0.2 C. At the current rate of 5 C, TiO_2 nanofiber-based electrodes exhibit a capacity retention as high as ~60%, irrespective of the presence of TiO_2 nanoglues in the electrode. The capacity of an electrode composed of TiO_2 nanofibers with nanoglues is fully recovered when it was measured at 0.2 C again. However, only ~89% of initial capacity is retained for TiO_2 nanofibers. This trend is similar to the results of cycle performance that the capacity of the TiO_2 nanofiber electrode decreases as a function of cycle number. Conventional TiO_2 anodes were prepared with 1D nanostructures, carbon conducting agents, and polymer binders, which exhibited the capacity retention of ~25% at 5 C compared to the capacity value of the initial current rate.[5,8,22] This comparison clearly shows that the nanoglue system offers favorable kinetics associated with electrons compared to the conventional electrode system. Recently, a TiO_2 anode prepared with hierarchical spheres consisting of TiO_2 nanosheets was reported to exhibit a superior capacity retention exceeding 80% at 5 C.[17] We expect that this binder-free process utilizing sol–gel nanoglues can produce high rate lithium ion batteries using such superior electrode materials.

Also, we compared the volumetric capacities of both the nanofiber electrode and nanofiber with the nanoglue electrode, which are shown in Table S1 (see ESI†). The nanofiber with the nanoglue electrode shows much higher volumetric capacity than that of the nanofiber electrode since the nanoglues can fill

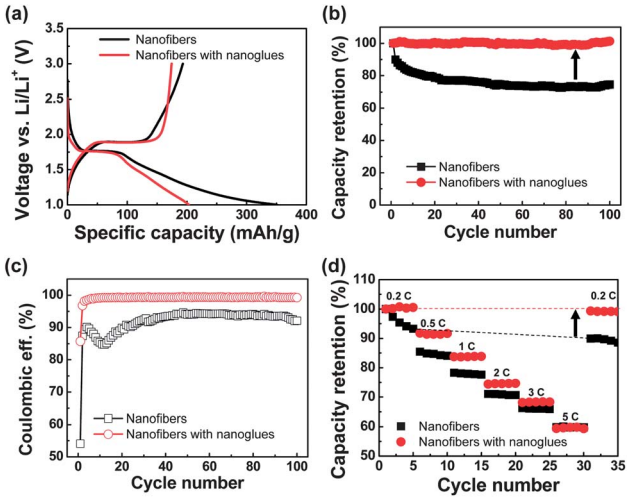

Fig. 3 Electrochemical performance of TiO_2 batteries: (a) 1st charge–discharge curves, (b) cycle performance, (c) Coulombic efficiency, and (d) rate capability.

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the huge volume of vacancies generated by the stacked nanofibers. The sol–gel nanoglues also have an advantage against the conventional electrodes that were prepared by mixing electrochemically active materials with carbon black and the polymer binder. The sol–gel nanoglues utilized in this demonstration are active materials. However, polymer binders are generally non-conductive and electrochemically inactive materials. In the voltage range of TiO_2 anode operation, carbon conducting agents are also electrochemically inactive. As a result, the presence of these conventional additives decreases the volumetric capacity of batteries.[23] This result indicates that our unique methodology using sol–gel nanoglues is beneficial to produce lithium ion batteries with high capacity density.

In order to interpret the electrochemical properties of the electrodes, electrochemical impedance spectroscopy (EIS) was conducted in the frequency range of 250 kHz to 10 mHz at an excitation amplitude of 5 mV and the results are shown in Fig. 4. Also, the equivalent circuit is presented to explain the electronic interfacial parameters of the TiO_2 electrode[8] as shown in the inset of Fig. 4. The Nyquist plots consist of a semicircle followed by a linear line. In general, the *x*-intercept of the semicircle corresponds to the Ohmic resistance between the electrolyte and electrode. The semicircle corresponds to the polarization resistance consisting of electrolyte transport resistance (R_{SEI}) in the solid electrolyte interface (SEI) layer and charge-transfer resistance (R_{ct}) of Li^+ in the electrode material. The linear region in the Nyquist plot is related to the diffusion of Li ions. Ohmic resistances are 1.9 and 2.6 Ω for the nanofiber electrode and nanofiber strengthened with the nanoglue electrode, respectively. The TiO_2 nanofiber electrode shows a high polarization resistance of ~400 Ω while the resistance significantly decreases below 100 Ω by applying TiO_2 sol–gel glues. Low polarization indicates generally the formation of a stable surface film and an efficient charge transfer of Li ions and electrons. Also, it was reported that the polarization resistance reflects the contact properties between electrode materials and the current collector as well as between electrode materials.[9] In this study, therefore, the lowest polarization resistance for the nanofiber electrode with nanoglues can be considered to be due to the increased contact area of TiO_2 and improved contact properties between TiO_2 nanofibers and the current collector due to the presence of nanoglues. For understanding the origin of kinetics further, the galvanostatic intermittent titration technique (GITT) was utilized and the results are shown in Fig. S6 (see ESI†). Fig. S6 (see ESI†) shows the GITT voltage profiles of both the TiO_2 nanofiber electrode and TiO_2 nanofiber with the nanoglue electrode during the first discharging and charging. From the obtained results, internal resistance for both electrodes was calculated as shown in Fig. S6c.† The nanofiber with the nanoglue electrode shows nearly one order of magnitude lower internal resistance than that of the nanofiber electrode, which indicates much improved kinetics associated with lithium for the nanofiber with the nanoglue electrode. This result is well consistent with the lower polarization resistance obtained from the EIS analysis. Therefore, it can be concluded that the utilization of TiO_2 sol–gel nanoglue improves the contact properties of TiO_2 nanofibers and provides the additional conducting pathways between TiO_2 nanofibers and the current collector, which leads to the enhanced electrochemical performance of TiO_2 nanofiber-based Li ion batteries.

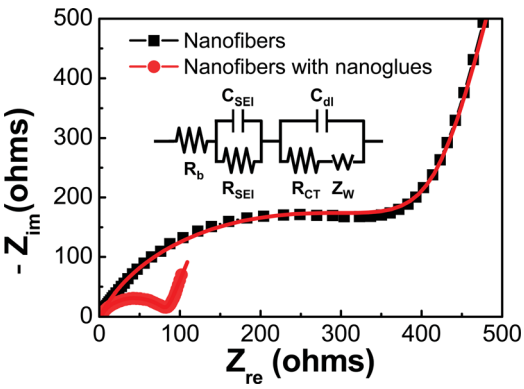

Fig. 4 Electrochemical impedance spectroscopy (EIS) analysis after the discharge–charge cycle.

In summary, we investigate the application of TiO_2 sol–gel nanoglues for the high performance TiO_2 nanofiber electrode without adding processing additives including polymeric binders and conducting additives. The electrode prepared with nanofibers strengthened with nanoglues shows excellent cycling performance up to 100 cycles with an initial Coulombic efficiency as high as 85.7%. Its electrochemical properties are higher than those of the directly electrospun TiO_2 nanofiber electrode. Nanoglues compensate the low initial Coulombic efficiency and unstable Coulombic efficiency with cycling that were observed in the nanofiber electrode system. The adhesion test reveals that the TiO_2 sol–gel nanoglues strengthen the adhesion strength of TiO_2 nanofibers. Thus, it can be concluded that the TiO_2 nanoglues provide enhanced contact properties of the nanofiber electrode, which is key for achieving excellent cycle performance with high initial Coulombic efficiency.

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